CSI4900 Honours Project – Fall 2026

DevSecOps-Oriented Real-Time Edge IoT Platform Using STM32/FreeRTOS, Raspberry Pi 4/QNX, and Ada/SPARK-Based Runtime Validation

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Project Period	September 9, 2026 – December 9, 2026

1. Project Overview

The objective of this project is to design, implement, and evaluate a small distributed real-time Edge/IoT platform combining embedded systems, real-time operating systems, edge computing, DevSecOps concepts, and formal runtime validation.

The system will consist of two main computing nodes:

- an STM32 Nucleo board running FreeRTOS, acting as the embedded sensing, control, and real-time task execution node;
- a Raspberry Pi 4 running QNX, acting as the supervisory edge computing platform responsible for data reception, coordination, monitoring, logging, and system-level analysis.

A lightweight Ada/SPARK-based validation component will be developed to formally specify and verify selected critical runtime rules. These may include message validity, sensor-data consistency, command authorization, range checking, or safe acceptance of data before processing by the QNX supervisory layer.

The project will also incorporate selected DevSecOps principles for embedded and real-time systems, including structured logging, runtime monitoring, communication validation, basic security checks, repeatable builds, testing, and systematic evaluation of system reliability.

The final system should demonstrate reliable communication between the embedded FreeRTOS node and the QNX edge platform under both normal and abnormal operating conditions.

2. Main Objectives

1. Develop real-time embedded software using STM32 and FreeRTOS.
2. Configure and develop applications for Raspberry Pi 4 using QNX.
3. Establish reliable communication between the STM32 and Raspberry Pi platforms.
4. Design a monitoring and logging framework for sensor data, system status, and control messages.
5. Introduce basic DevSecOps practices into the embedded-system development workflow.
6. Develop a small Ada/SPARK component for formal validation of selected critical data or communication rules.
7. Evaluate timing, responsiveness, reliability, and fault-handling behavior.
8. Document and present the complete system architecture, implementation, experiments, and results.

3. Semester Work Plan

Week 1 — September 9: Project Definition and Requirements

- Review the complete project scope with the supervisor.
- Confirm hardware and software requirements.
- Define the system architecture.
- Identify the STM32 board, Raspberry Pi 4, sensors, and communication interfaces.
- Define functional and non-functional requirements.
- Define preliminary performance criteria.

Expected output: Project requirements document and preliminary system architecture.

Week 2 — September 16: Background Study and Development Environment

- Review FreeRTOS architecture and scheduling concepts.
- Review QNX architecture and real-time features.
- Review communication options between STM32 and Raspberry Pi.
- Review Ada/SPARK and formal verification concepts.
- Install and configure the required development environments.
- Establish source-code version control.

Expected output: Development environment operational and short technical literature/background summary.

Week 3 — September 23: STM32/FreeRTOS Initial Implementation

- Configure the STM32 development board.
- Create the initial FreeRTOS application.
- Implement multiple real-time tasks.
- Test task priorities, timing, and synchronization.
- Implement basic sensor or simulated-data acquisition.

Expected output: Working STM32/FreeRTOS prototype with multiple real-time tasks.

Week 4 — September 30: Raspberry Pi 4/QNX Platform Setup

- Configure QNX on the Raspberry Pi 4.
- Develop a basic QNX application.
- Test processes, threads, timing, and system services.
- Implement an initial data-reception application.

Expected output: Operational Raspberry Pi/QNX development platform.

Week 5 — October 7: Communication Interface Design

- Select and configure the communication protocol between STM32 and Raspberry Pi.
- Define the message structure.
- Define sensor-data, status, and command-message formats.
- Implement initial communication between both platforms.
- Possible mechanisms include UART, Ethernet, TCP/UDP, or another suitable interface.

Expected output: Basic bidirectional communication between STM32/FreeRTOS and Raspberry Pi/QNX.

Week 6 — October 14: Reliable Data Exchange

- Improve the communication framework.
- Add timestamps, message identifiers, and basic integrity checks.
- Implement error detection.
- Handle lost, invalid, delayed, or duplicated messages.
- Test communication under different operating conditions.

Expected output: Reliable communication and message-handling framework.

Week 7 — October 21: Monitoring and Structured Logging

- Implement structured logging on the QNX platform.
- Record sensor values, system status, communication events, and errors.
- Develop basic runtime monitoring.
- Establish metrics for system responsiveness and reliability.

Expected output: Functional monitoring and logging subsystem.

Week 8 — October 28: DevSecOps-Oriented Integration

- Introduce automated or repeatable build procedures.
- Organize source-code management and versioning.
- Define basic test procedures.
- Add communication validation and basic security checks.
- Document the development and testing workflow.

Expected output: Documented lightweight DevSecOps workflow for the embedded/edge system.

Week 9 — November 4: Ada/SPARK Validation Component – Design

- Select one critical system function for formal validation.
- Define its requirements and safety properties.
- Implement an initial Ada/SPARK component.
- Possible targets include message validity, sensor-value range checking, data consistency, command authorization, communication-state validation, and safe acceptance/rejection of received data.

Expected output: Initial Ada/SPARK validation module and formal specification.

Week 10 — November 11: Ada/SPARK Verification and Integration

- Apply SPARK verification tools to the selected component.
- Address verification warnings or errors.
- Integrate the validation component with the rest of the system.
- Test valid and invalid inputs.

Expected output: Verified Ada/SPARK component integrated into the system.

Week 11 — November 18: Complete System Integration

- Integrate STM32/FreeRTOS, communication, Raspberry Pi/QNX, monitoring, and Ada/SPARK validation.
- Perform end-to-end functional testing.
- Correct integration problems.
- Finalize the core prototype.

Expected output: Complete working system prototype.

Week 12 — November 25: Experimental Evaluation

- Conduct systematic experiments to evaluate communication latency, response time, timing behavior, task execution, reliability, message-loss handling, invalid-data handling, and abnormal-condition behavior.
- Collect and organize experimental results.

Expected output: Experimental dataset, tables, graphs, and preliminary analysis.

Week 13 — December 2: Final Analysis and Report Preparation

- Analyze experimental results.
- Compare results with initial project requirements.
- Discuss system limitations.
- Identify possible future improvements.
- Complete the technical documentation.
- Prepare the final report.
- Prepare the final presentation.

Expected output: Draft final report and presentation slides.

Week 14 — December 9: Final Demonstration and Presentation

- Present the completed project and demonstrate the prototype.
- Deliver an approximately 15-minute presentation, followed by questions and discussion.
- Cover the problem and motivation, objectives, architecture, implementation, communication framework, DevSecOps approach, Ada/SPARK validation, methodology, results, demonstration, conclusions, and future work.

Expected output: Working prototype, source code, technical documentation, experimental results, final written report, and 15-minute oral presentation.

4. Final Project Deliverables

1. Working Prototype

A functional distributed system demonstrating communication between STM32/FreeRTOS and Raspberry Pi 4/QNX, with monitoring and runtime validation.

2. Source Code

All source code should be clearly organized and documented, including STM32/FreeRTOS software, QNX software, communication modules, monitoring and logging components, Ada/SPARK validation component, and testing scripts or utilities where applicable.

3. Experimental Evaluation

The student must perform experiments evaluating system performance and reliability. Results should include quantitative measurements where possible, including latency, timing, responsiveness, reliability, communication errors, and behavior under normal and abnormal conditions.

4. Final Report

The final report should describe the introduction and motivation, background and related technologies, requirements, system architecture, design methodology, implementation, FreeRTOS subsystem, QNX subsystem, communication architecture, DevSecOps approach, Ada/SPARK validation, experimental methodology, results and analysis, limitations, conclusions, future work, and references.

5. Final Oral Presentation

The student will deliver a 15-minute final presentation summarizing the complete project. The presentation should emphasize the engineering work performed, technical challenges, implemented solutions, experimental results, and lessons learned.

5. Optional Extension Goals

If the core system is completed successfully and time permits, the student may investigate one optional extension, such as:

spark on stm32

- secure communication
- message authentication
- enhanced fault detection
- automated testing
- continuous integration
- remote monitoring dashboard
- lightweight anomaly detection
- small on-device AI functionality

These features are considered optional stretch goals and should only be attempted after the main project requirements have been successfully completed.

6. Expected Final Outcome

At the end of the semester, the project should demonstrate a complete small-scale real-time Edge/IoT system integrating:

- **STM32 + FreeRTOS → Real-Time Embedded Node**
- **Raspberry Pi 4 + QNX → Supervisory Edge Platform**
- **Monitoring + Logging → Runtime Observability**
- **Ada/SPARK → Critical Runtime Validation**
- **DevSecOps Principles → Structured Development, Testing, and Reliability**

The project should demonstrate not only that the system operates correctly, but also that its timing behavior, communication reliability, and selected critical runtime properties have been systematically evaluated.

project is a research and engineering prototype focused on integrating real-time control, formal validation, networking, and fail-safe behaviour in one testbed. I am evaluating the architecture and its behaviour under normal and abnormal conditions.”